

# Parameters & Units

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All physical quantities in the codebase use the same unit system: **ms for time**, **mV for voltage**, **nF for capacitance**,  **$\mu\text{S}$  for conductance**, **nA for current**, **Hz for rates**. Time fields carry an explicit `_ms` suffix (`sim_ms`, `ref_ms_E`, `tau_gaba`); CLI flags follow the same convention (`-t-ms 600`). A  $\Delta t$  of 1 means 1 ms, not 1 s.

## Quantities

| Quantity                 | Unit          | Typical value                | Variable                                       |
|--------------------------|---------------|------------------------------|------------------------------------------------|
| Integration step         | ms            | 0.25                         | <code>dt</code> , <code>DT_MS</code>           |
| Simulation length        | ms            | 600                          | <code>sim_ms</code>                            |
| Membrane time constant   | ms            | 20 (E), 5 (I)                | <code>tau_m_E</code> , <code>tau_m_I</code>    |
| Refractory period        | ms            | 3 (E), 1.5 (I)               | <code>ref_ms_E</code> , <code>ref_ms_I</code>  |
| AMPA decay               | ms            | 2                            | <code>tau_ampa</code>                          |
| GABA decay               | ms            | 9                            | <code>tau_gaba</code>                          |
| Resting / leak potential | mV            | −65                          | <code>E_L</code>                               |
| Spike threshold          | mV            | −50                          | <code>V_th</code>                              |
| Reset potential          | mV            | −65                          | <code>V_reset</code>                           |
| AMPA reversal            | mV            | 0                            | <code>E_e</code>                               |
| GABA reversal            | mV            | −80                          | <code>E_i</code>                               |
| Membrane capacitance     | nF            | 1.0 (E), 0.5 (I)             | <code>C_m_E</code> , <code>C_m_I</code>        |
| Leak conductance         | $\mu\text{S}$ | 0.05 (E), 0.1 (I)            | <code>g_L_E</code> , <code>g_L_I</code>        |
| External drive           | $\mu\text{S}$ | 0.0006 (async), 0.003 (PING) | <code>t_e_async</code> , <code>t_e_ping</code> |
| Input current (CUBA)     | nA            | 20 per spike                 | <code>input_scale</code>                       |
| CUBA weight std          | nA            | 32                           | <code>W_STD_CUBA</code>                        |
| Max input rate           | Hz            | 25                           | <code>max_rate_hz</code>                       |
| Population firing rate   | Hz            | 20–80                        | <code>r_E</code> , <code>r_I</code>            |
| Gamma frequency          | Hz            | 30–80                        | <code>f_0</code>                               |

## COBA / PING biophysical constants

Used by COBA and PING. Values follow neuroscience conventions (cf. Dayan & Abbott, Gerstner *Neuronal Dynamics*); the E:I asymmetry in  $\tau_m, C_m, g_L, \tau_{\text{ref}}$  produces the timescale separation that makes PING dynamics possible.

| Parameter                | E population | I population |
|--------------------------|--------------|--------------|
| $\tau_m$ (ms)            | 20           | 5            |
| $C_m$ (nF)               | 1.0          | 0.5          |
| $g_L$ ( $\mu\text{S}$ )  | 0.05         | 0.1          |
| $\tau_{\text{ref}}$ (ms) | 3            | 1.5          |
| $E_L$ (mV)               | −65          | −65          |
| $V_{\text{th}}$ (mV)     | −50          | −50          |
| $V_{\text{reset}}$ (mV)  | −65          | −65          |

|                      |     |     |
|----------------------|-----|-----|
| $E_e$ (mV, reversal) | 0   | 0   |
| $E_i$ (mV, reversal) | -80 | -80 |

Synapse time constants:  $\tau_{\text{AMPA}} = 2$  ms (excitation),  $\tau_{\text{GABA}} = 9$  ms (inhibition). These set the ceiling on PING's  $\Delta t$ -stability: once  $\Delta t \gtrsim \tau_{\text{GABA}}$  the  $E \rightarrow I \rightarrow E$  loop cannot complete within one step.

## Internal consistency

The chosen units are self-consistent — no conversion factors appear in the integration code. Two equations carry the whole system.

The membrane time constant is  $\tau = C/g$ . With  $C$  in nF and  $g$  in  $\mu\text{S}$ ,

$$\tau_{[\text{ms}]} = \frac{C_{[\text{nF}]}}{g_{[\mu\text{S}]}}$$

so  $C_m = 1$  nF and  $g_L = 0.05$   $\mu\text{S}$  give  $\tau_m = 20$  ms directly.

The LIF voltage update is  $dv = (\Delta t/C)(-g_L(v - E_L) + I)$ . With  $\Delta t$  in ms,  $C$  in nF,  $v, E$  in mV,  $g$  in  $\mu\text{S}$ , and  $I$  in nA,

$$dv_{[\text{mV}]} = \frac{\Delta t_{[\text{ms}]}}{C_{[\text{nF}]}} \cdot I_{[\text{nA}]}$$

because  $\text{ms} \cdot \text{nA} / \text{nF} = \text{mV}$  exactly.

Conductance-current products share the same ledger:  $g(v - E)$  is  $\mu\text{S} \times \text{mV} = \text{nA}$ , so synaptic currents fold into  $I$  alongside any direct input current without a scale factor.

## Why not SI?

Pure SI (F, S, V, A, s) forces every value to a large negative exponent —  $C_m = 10^{-9}$  F,  $g_L = 5 \times 10^{-8}$  S,  $\Delta t = 2.5 \times 10^{-4}$  s. The neuroscience convention (ms, mV, nF,  $\mu\text{S}$ , nA) keeps every typical value between  $10^{-3}$  and  $10^2$ , which makes numerical debugging and human intuition faster. The tradeoff is that readers have to trust the unit consistency rather than verify it by plugging into SI formulas — this page is here so that trust is auditable.